

Thomas Stoll

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Summary

Research fellow with over 7 years of experience in auditory neuroscience, specializing in the development of objective diagnostic tools and investigating auditory function through electrophysiology and behavioral measures. Skilled in EEG acquisition, Python-based data science, computational modeling, custom equipment manufacturing, and signal processing. Committed to improving patient outcomes by integrating deep learning models into real-world auditory diagnostics and therapeutics.

Education

University of Rochester PhD, Biomedical Engineering Maddox Lab	2018 – 2024
University of Rochester MS, Biomedical Engineering Maddox Lab	2018 – 2019
University at Buffalo BS, Biomedical Engineering Summa cum laude	2014 – 2018
University at Buffalo BA, Mathematics Summa cum laude	2014 – 2018

Skills

- **Computational Modeling & AI:** Deep Learning (CNNs, Encoder-Decoder architectures, time-series signals), Predictive Modeling, Linear Mixed-Effects (LME) Modeling
- **Programming & Software:** Python (NumPy, Pandas, Matplotlib, MNE-python), MATLAB, Shell Scripting/SSH, Version Control (Git)
- **Electrophysiology & Data Acquisition:** EEG, Auditory Brainstem Response (ABR/pABR) collection and analysis, Behavioral Assessments, Signal Processing
- **Hardware & Prototyping:** 3D Design & Modeling (CAD), 3D Printing (FDM), Custom Equipment Fabrication (Machining, woodworking, electronics), Soldering

Experience

Postdoctoral research fellow , University of Michigan Maddox lab	2024 – current
• Prepared datasets by arranging data from past experiments in a standardized format. • Designed, trained, and tested different architectures, hyperparameters, and loss functions. • Assisted in administrative tasks, regulatory compliance. • Mentored new lab members, assisted collaborators in EEG setup and analysis.	

Participant, Telluride Neuromorphic Cognition Engineering Workshop July 2025

- Led a project investigating neural signatures of attention switching with EEG.
- Co-led a project focused on reconstructing speech signals from imagined speech using an iEEG dataset.
- Assisted in projects employing real-time decoding of attention through EEG to control hardware and software for demonstrations.
- Assisted in equipment setup and validation, provided data processing pipelines and aided in integration of analysis methods for different projects.

PhD researcher, University of Rochester 2018 – 2024
Maddox lab

Dissertation: Advancements in Stimulus and Analysis Paradigms to Study Subcortical Sound Encoding

- Conducted experiments investigating the place specificity of the parallel auditory brainstem response through both computational models and human electrophysiology recordings.
- Investigated effects of selective auditory attention throughout the auditory pathway. Assisted in developing tools to extend the temporal response function framework to the subcortex. Fabricated lab equipment and electrodes to record signals from the tympanic membrane.
- Assisted in regulatory compliance, mentored new lab members.

Teaching Assistant, University of Rochester 2019 – 2020
Biomedical Engineering Department

- Biomedical Signals, Systems, and Imaging: Held office hours, led hands-on lab sections, created rubrics, and graded lab reports.
- Biomedical Computation and Statistics: Held office hours, led computer labs, graded assignments.

Research assistant, Toshiba Stroke and Vasculature Research Center 2015 – 2017
Ionita Lab

- Created and refined 3D models of patient vasculature, printed patient-specific phantoms
- Developed methodology to test and analyze 3D printed materials to determine combinations of materials necessary to best simulate vasculature for testing new devices or training medical students.

Publications

Tabaczynski, J. R., **Stoll, T.**, Shepard, L., Siddiqui, M. I., Karkhanis, N. V., Sommer, K., ... & Ionita, C. N. (2018). "Use of patient specific 3D printed (3DP) neurovascular phantoms for mechanical assessment of devices used in image guided minimally invasive procedures." In *Medical Imaging 2018: Imaging Informatics for Healthcare, Research, and Applications* (Vol. 10579, pp. 117-125). SPIE.

Stoll, T. J., & Maddox, R. K. (2023). Enhanced Place Specificity of the Parallel Auditory Brainstem Response: A Modeling Study. In *Trends in Hearing*, 2023; 27, <https://doi.org/10.1177/23312165231205719>.

Stoll, T. J., & Maddox, R. K. (2024). Enhanced Place Specificity of the Parallel Auditory Brainstem Response: An Electrophysiological Study. In *JARO* **25**, 477–489 (2024). <https://doi.org/10.1007/s10162-024-00959-w>

Stoll, T. J., Vandjelovic, N. D., Polonenko, M. J., Li, N. R., Lee, A. K., & Maddox, R. K. (2025). The auditory brainstem response to natural speech is not affected by selective attention. *PLoS biology*, 23(10), e3003407. <https://doi.org/10.1371/journal.pbio.3003407>

Patents

Maddox, R. & **Stoll, T.** (2026) “SYSTEMS AND METHODS FOR PREDICTING INDIVIDUAL BRAIN RESPONSES TO ACOUSTIC STIMULI USING DEEP NEURAL NETWORKS” U.S. Provisional Patent Application 63/986,368, filed February 19, 2026. Patent Pending

Presentations

Stoll, T. & Maddox, R. (2020, January). “Modeling Improved Place Specificity in the Parallel Auditory Brainstem Response.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, San Jose, CA.

Stoll, T. & Maddox, R. (2022, February). “Place Specificity of the Parallel Auditory Brainstem Response.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, Virtual.

Stoll, T. & Maddox, R. (2022, July). “Place Specificity of the Parallel Auditory Brainstem Response.” Poster session presented at the biennial meeting of the *Auditory System Gordon Research Seminar and Conference*, Providence, RI.

Stoll, T. & Maddox, R. (2023, February). “Investigating the Effect of Selective Attention on Speech Encoding from Auditory Nerve to Cortex.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, Orlando, FL.

Stoll, T. & Maddox, R. (2024, February). “Exploring a Possible EEG Marker of Anticipatory Processing in a Natural Speech-Based Selective Attention Task.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, Anaheim, CA.

Stoll, T., Casale J., & Maddox, R. (2025, February). “Towards a Generalized Deep Neural Network Model of Human EEG Responses to Sounds.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, Orlando, FL.

Stoll, T. & Maddox, R. (2026, February). “The Effect of Electrode Placement on the Auditory Brainstem Response.” Poster session presented at the annual meeting of the *Association for Research in Otolaryngology*, San Juan, PR.

Honors & Awards

University at Buffalo Dean’s List	2014 – 2018
MidWinter Meeting Travel Award, Association for Research in Otolaryngology	2020
MidWinter Meeting Travel Award, Association for Research in Otolaryngology	2022